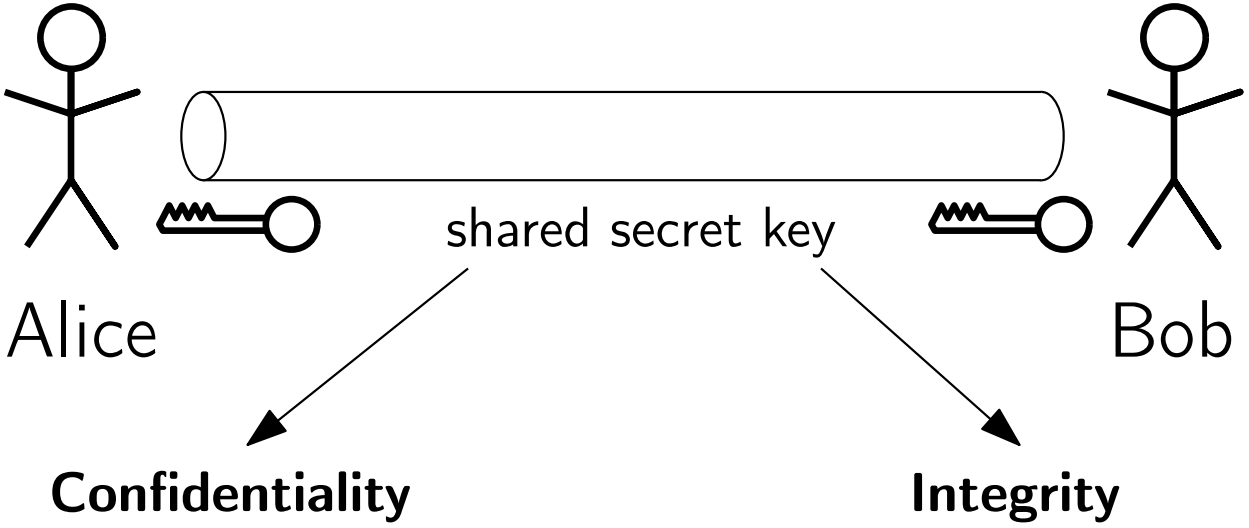
IY3660 Applications of Cryptography Guest Lecture
Quantum Cryptography

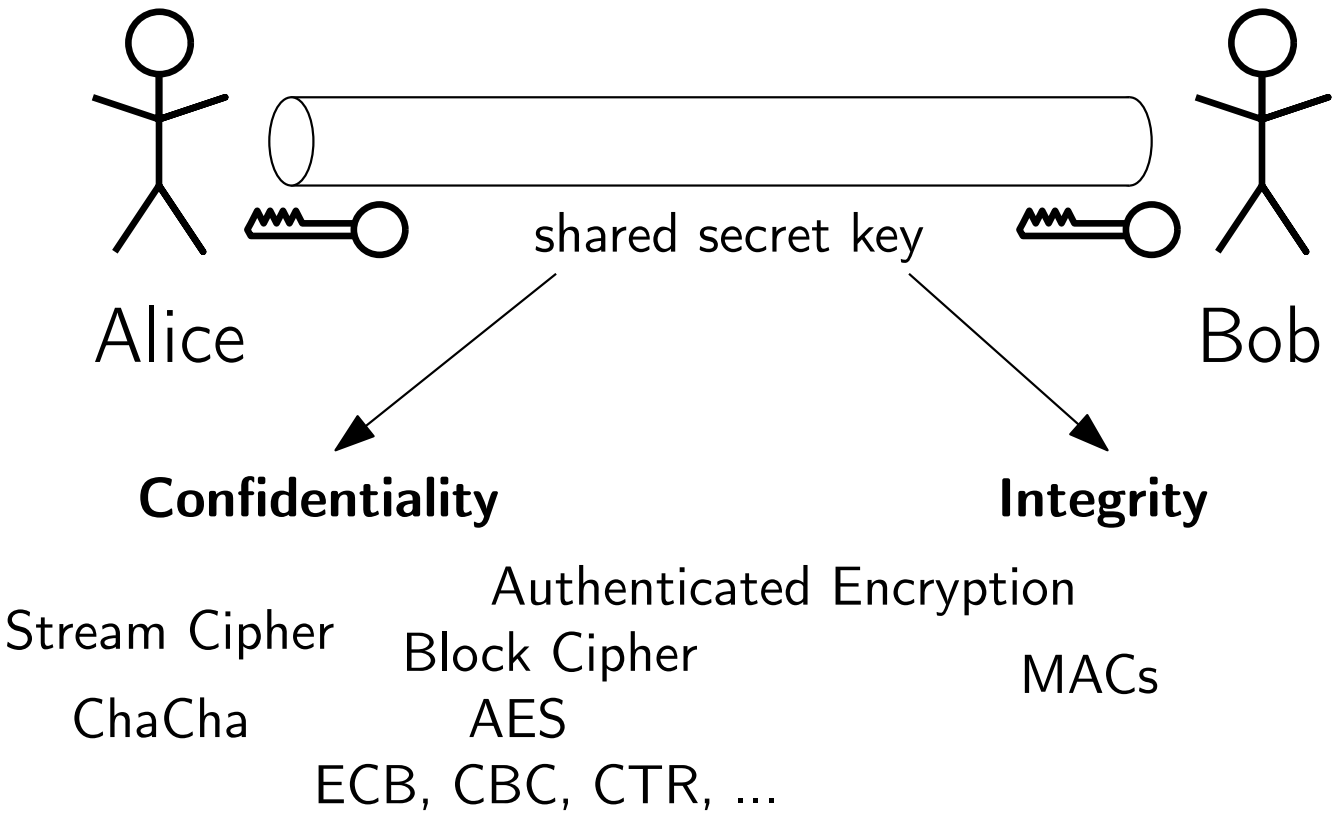
Simon Pohmann

6 Nov 2025

Symmetric Crypto

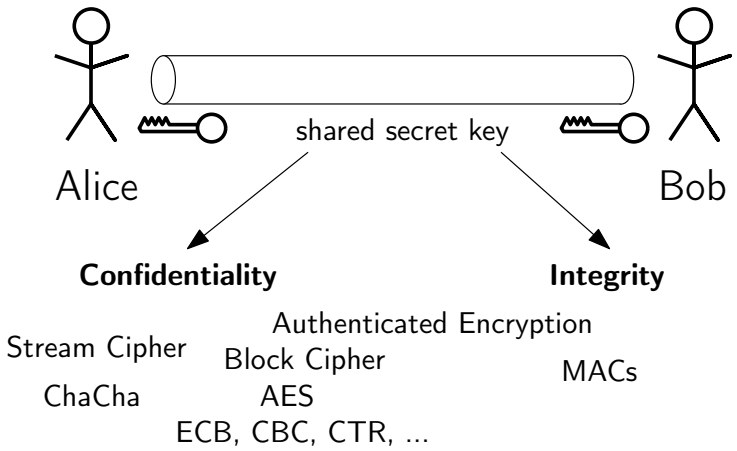


Symmetric Crypto Challenge: Key Exchange



But how do Alice and Bob get a shared key?

Symmetric Crypto Challenge: Key Exchange



But how do Alice and Bob get a shared key?

- Next Lecture(s): **Asymmetric Cryptography**
- Today: **Quantum Cryptography**

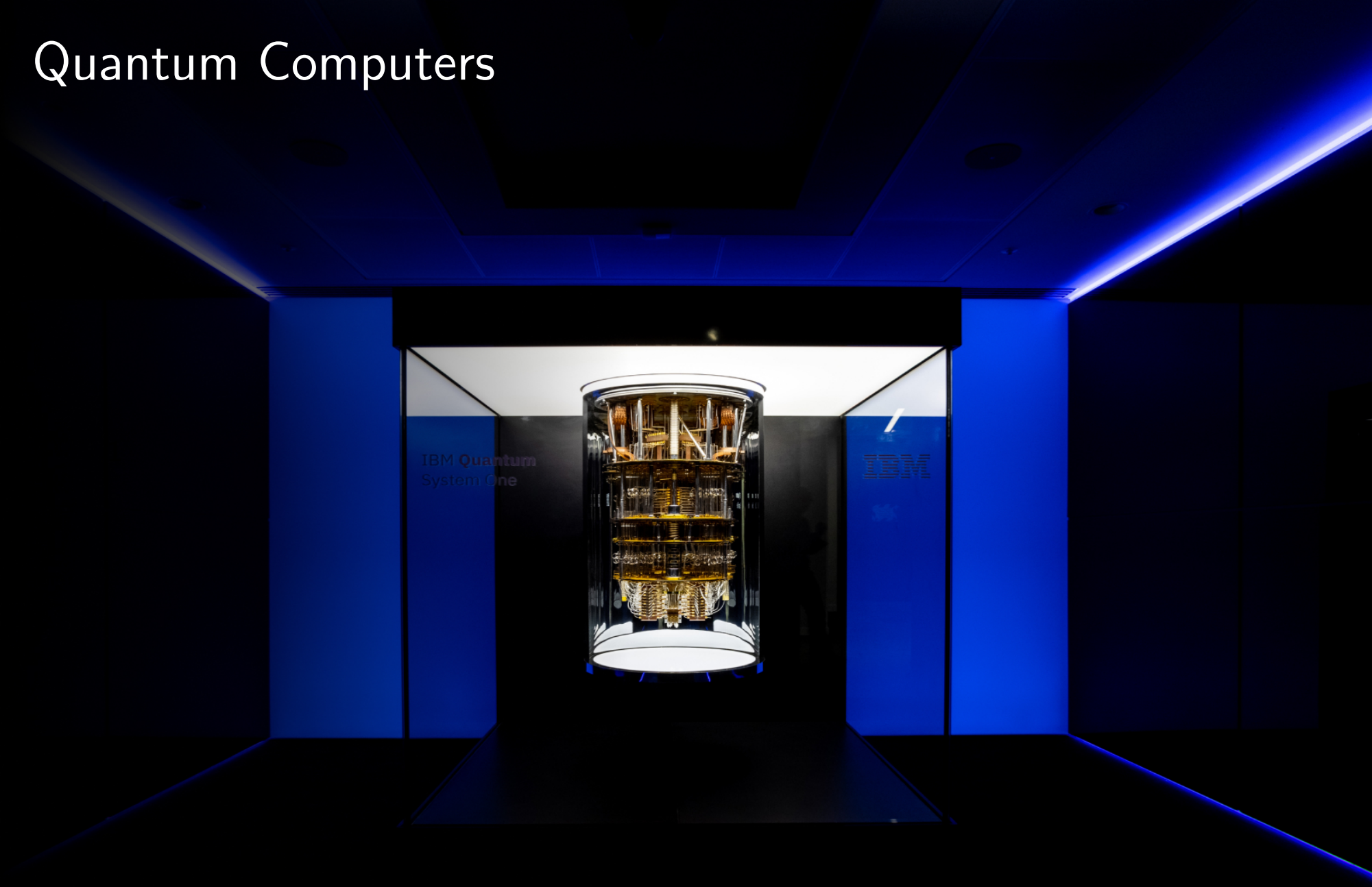
Quantum Cryptography

Not to confuse with **Post-Quantum Cryptography**

The adversary has a quantum computer
$\rightarrow$ Future Lecture

Alice and Bob use a quantum computer
$\rightarrow$ Today

Quantum Computers



IBM Quantum System 1, from https://www.ibm.com/quantum/systems/feature_card_1.jpg

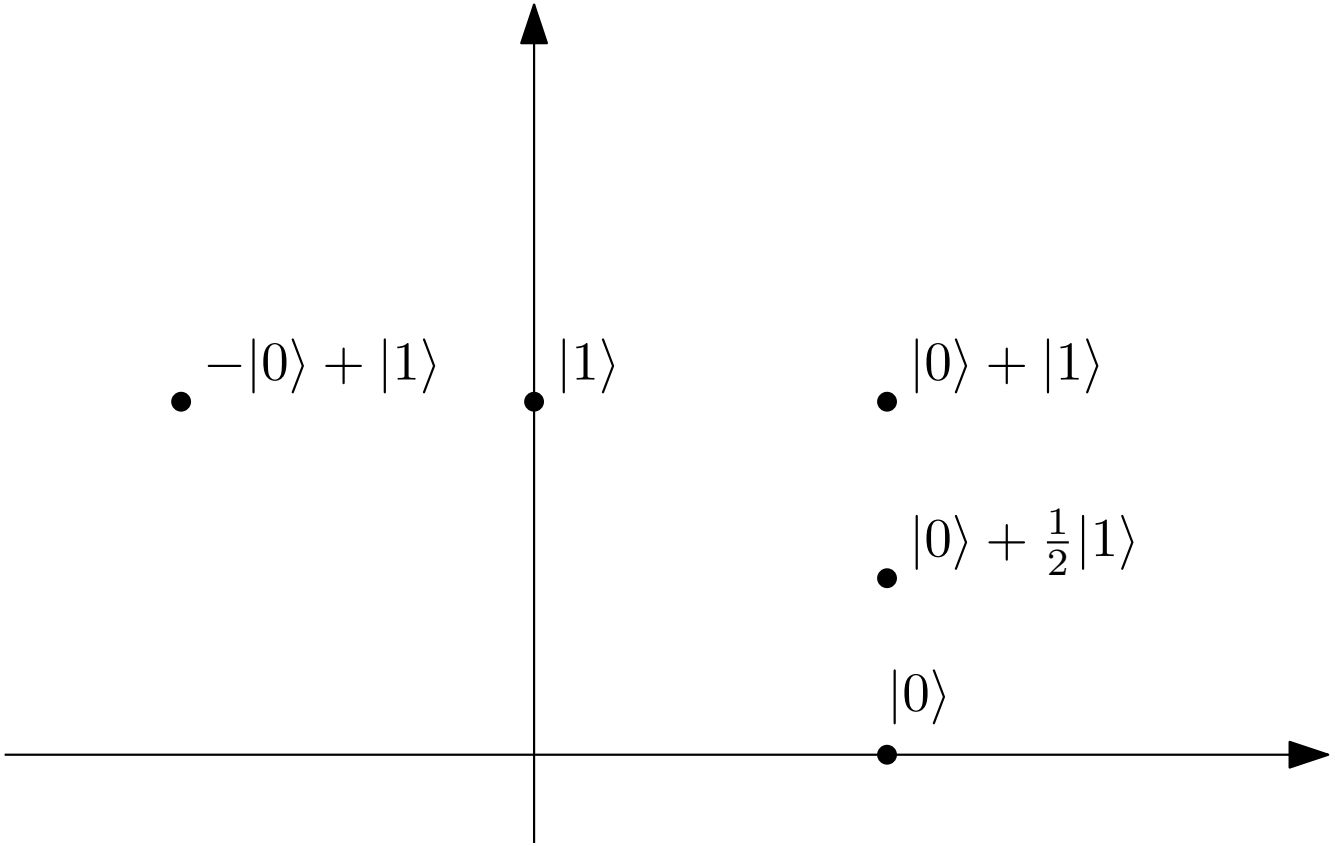
History of quantum computers

1980	Idea of Quantum Computer
1985	Deutsch's algorithm
1991	Quantum Key Distribution
1994	Shor's algorithm
2000	First "Quantum Computer" (3 qubits)
2019	"Quantum Supremacy" (54 qubits)
2023	IBM Condor: 1121 qubits
2025	Team at Caltech: 6100 qubits

The qubit

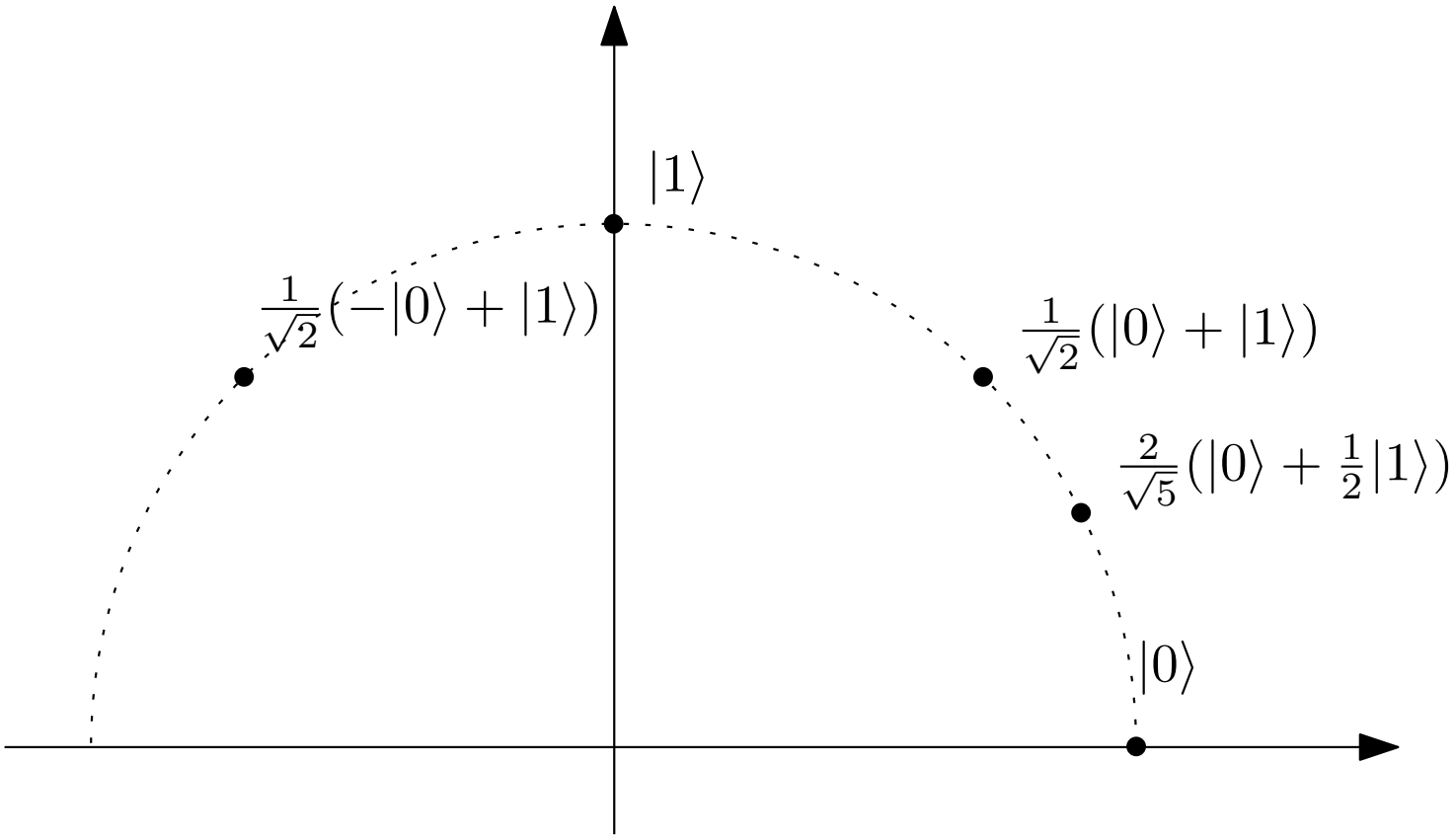
	Classical bit	Qubit
State	0 or 1	$\mathbb{C}$-vector space spanned by $\|0\rangle, \|1\rangle$
Implementation	low/high voltage	Quantum phenomena e.g. spin, polarization, ...
Operations	AND, XOR, ...	Transform, Measurement

Example



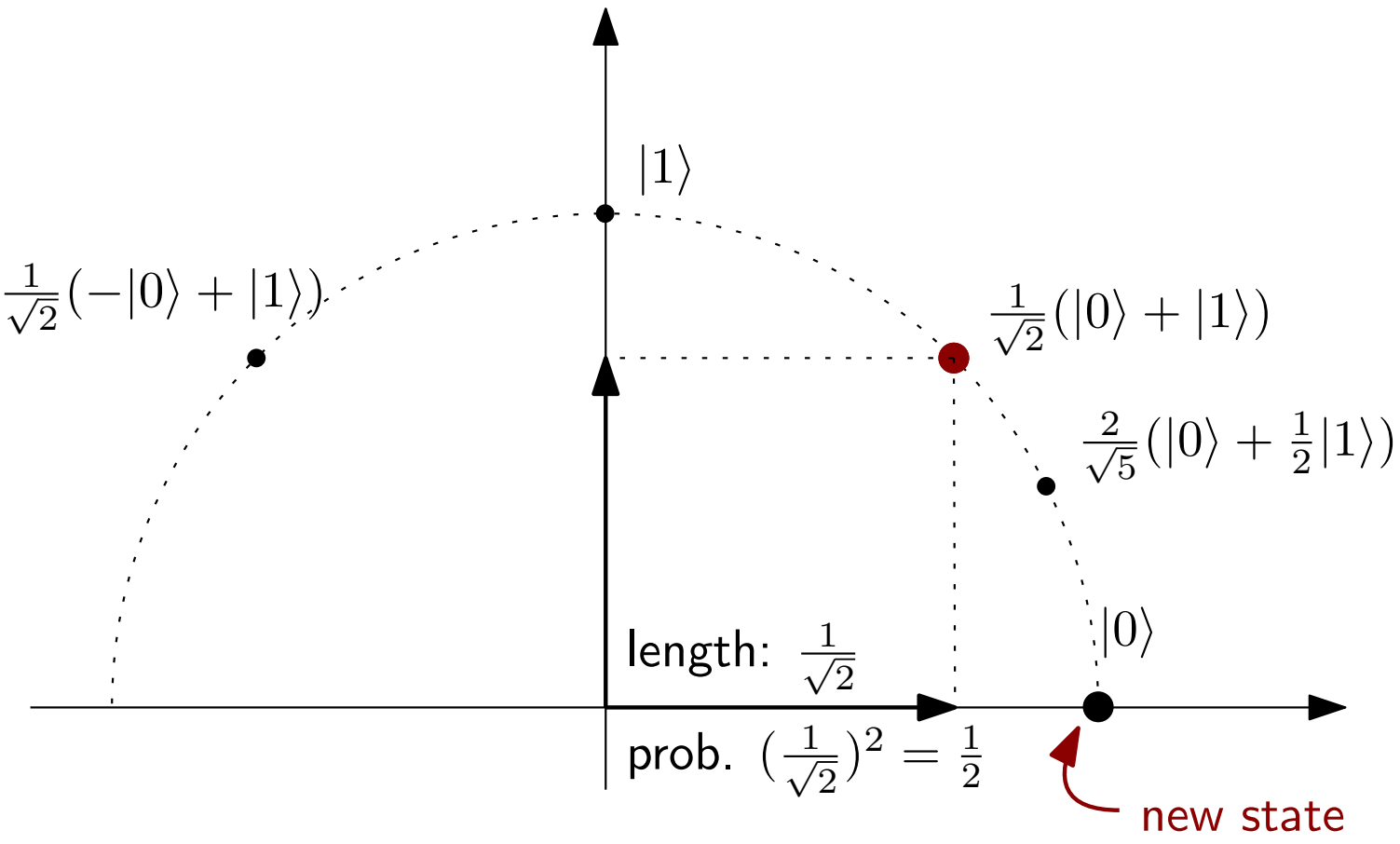
We work over $\mathbb{C}$, so also $|0\rangle + i|1\rangle$ exist...

Example

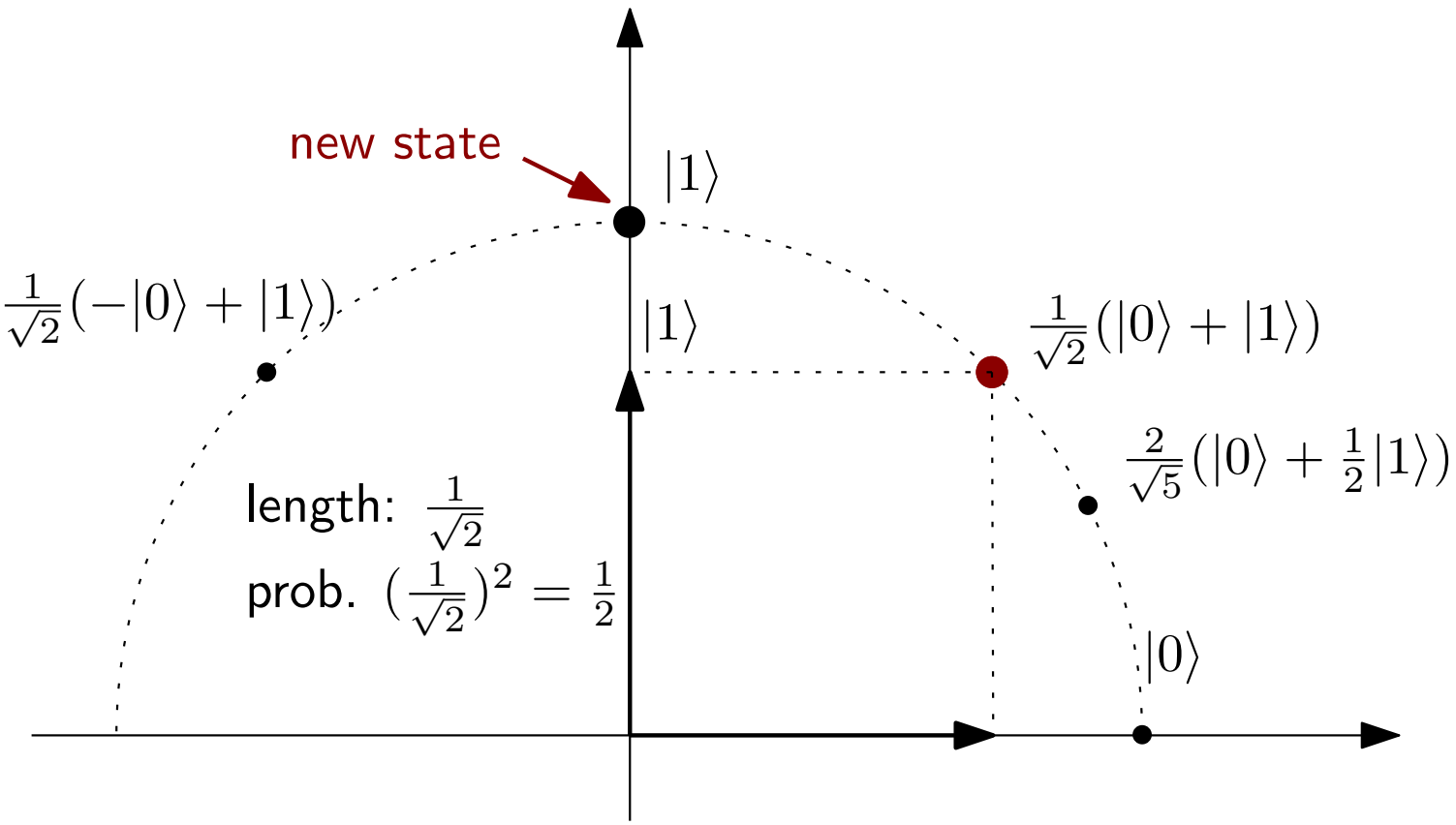


We work over $\mathbb{C}$, so also $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ exist...

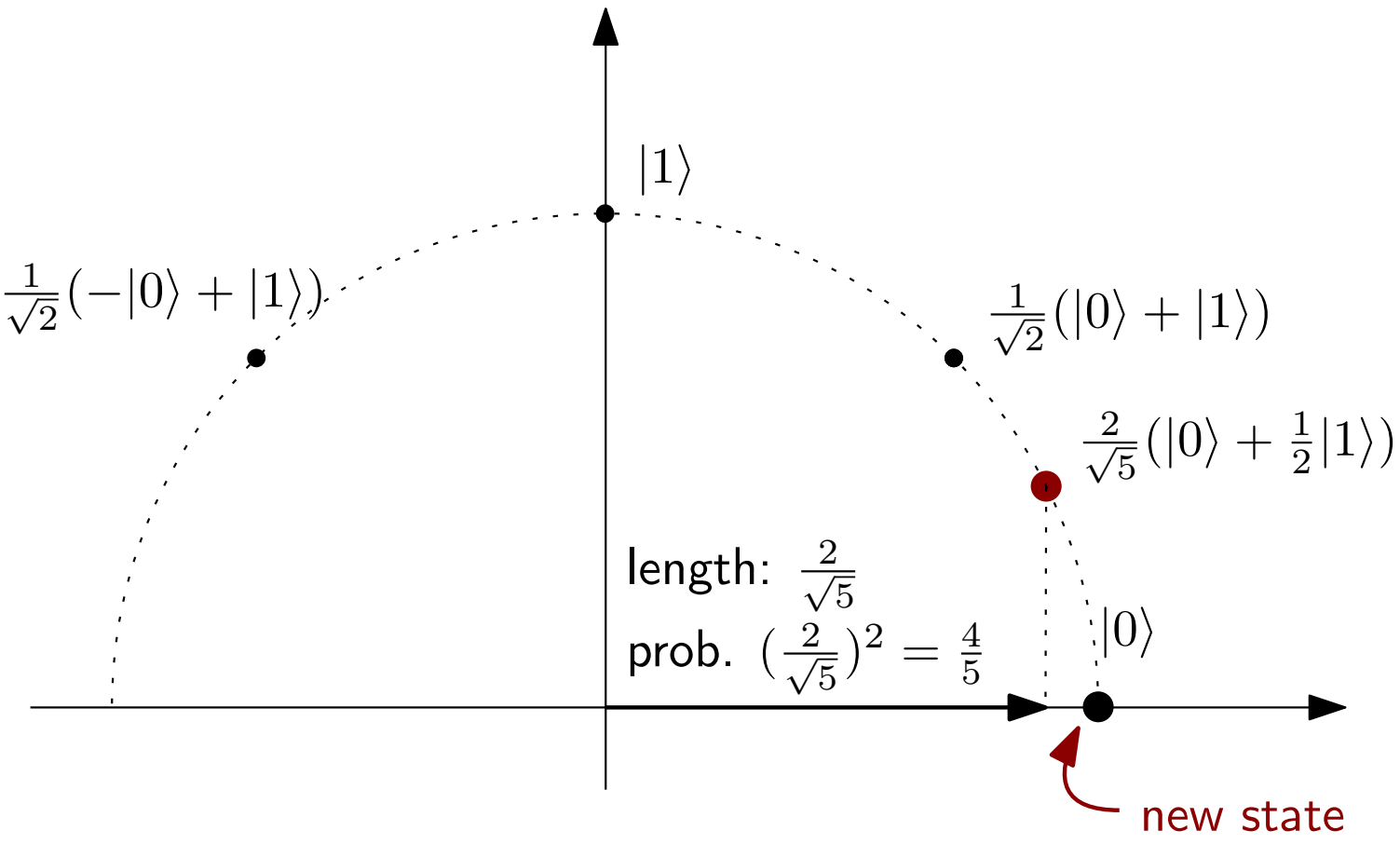
Measurement



Measurement



Measurement



The no-cloning theorem

You are given a qubit in one of four states

$$|0\rangle, |1\rangle, |0\rangle + |1\rangle, |0\rangle - |1\rangle$$

Can you find out which state it is?

No!

Say we measure and get 1

- Original state could be $|1\rangle, |0\rangle + |1\rangle$ or $|0\rangle - |1\rangle$
- But new state is $|1\rangle$

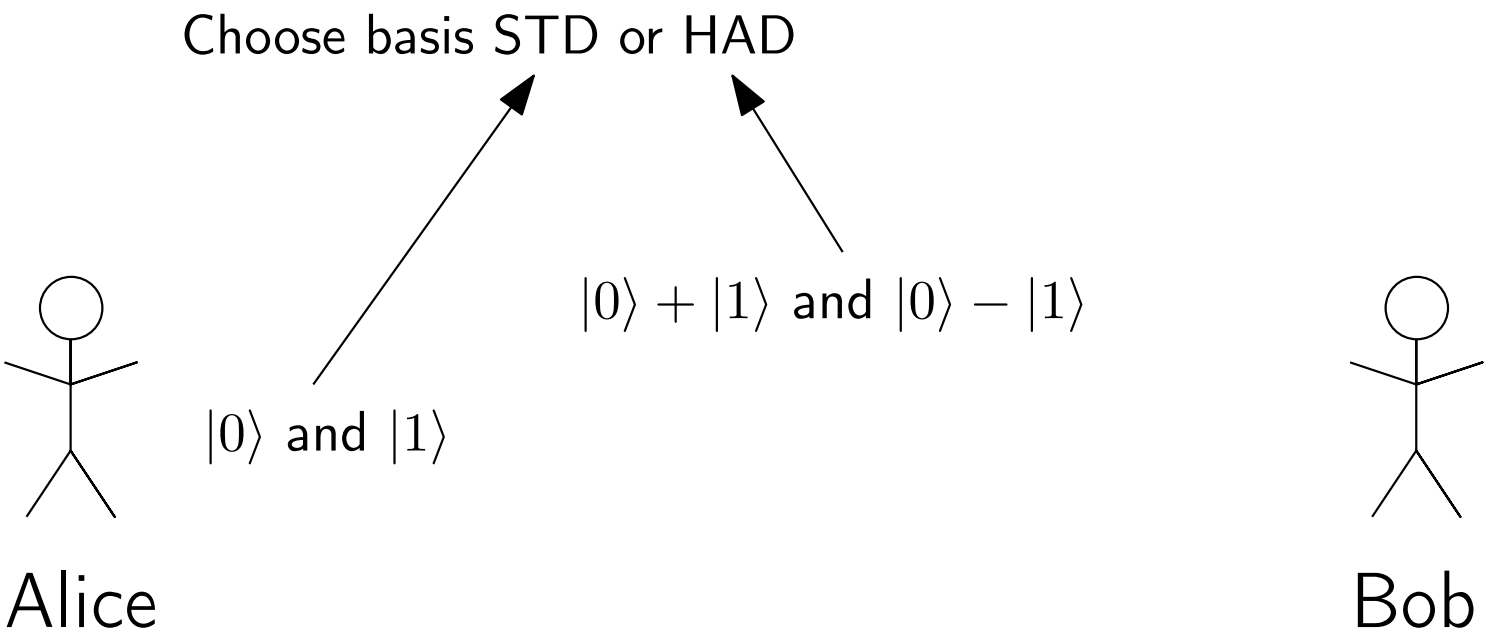
Application - Generating truly random numbers

- Have qubit $|0\rangle$
- Apply *Hadamard-transform*

$$H: \quad \begin{array}{l} |0\rangle \mapsto |0\rangle + |1\rangle \\ |1\rangle \mapsto |0\rangle - |1\rangle \end{array}$$

- Measure!
- The result is 0 or 1, with probability 50%

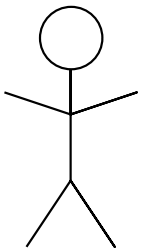
Quantum Key Distribution



Quantum Key Distribution

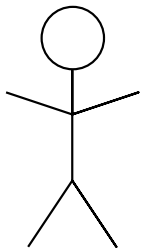
Choose basis STD or HAD
Choose α as 0 or 1

If STD: $|b\rangle = |\alpha\rangle$
If HAD: $|b\rangle = H|\alpha\rangle$



Alice

Hadamard transform
$|0\rangle \mapsto |0\rangle + |1\rangle$
$|1\rangle \mapsto |0\rangle - |1\rangle$



Bob

Quantum Key Distribution

Choose basis STD or HAD
Choose α as 0 or 1

If STD: $|b\rangle = |\alpha\rangle$
If HAD: $|b\rangle = H|\alpha\rangle$

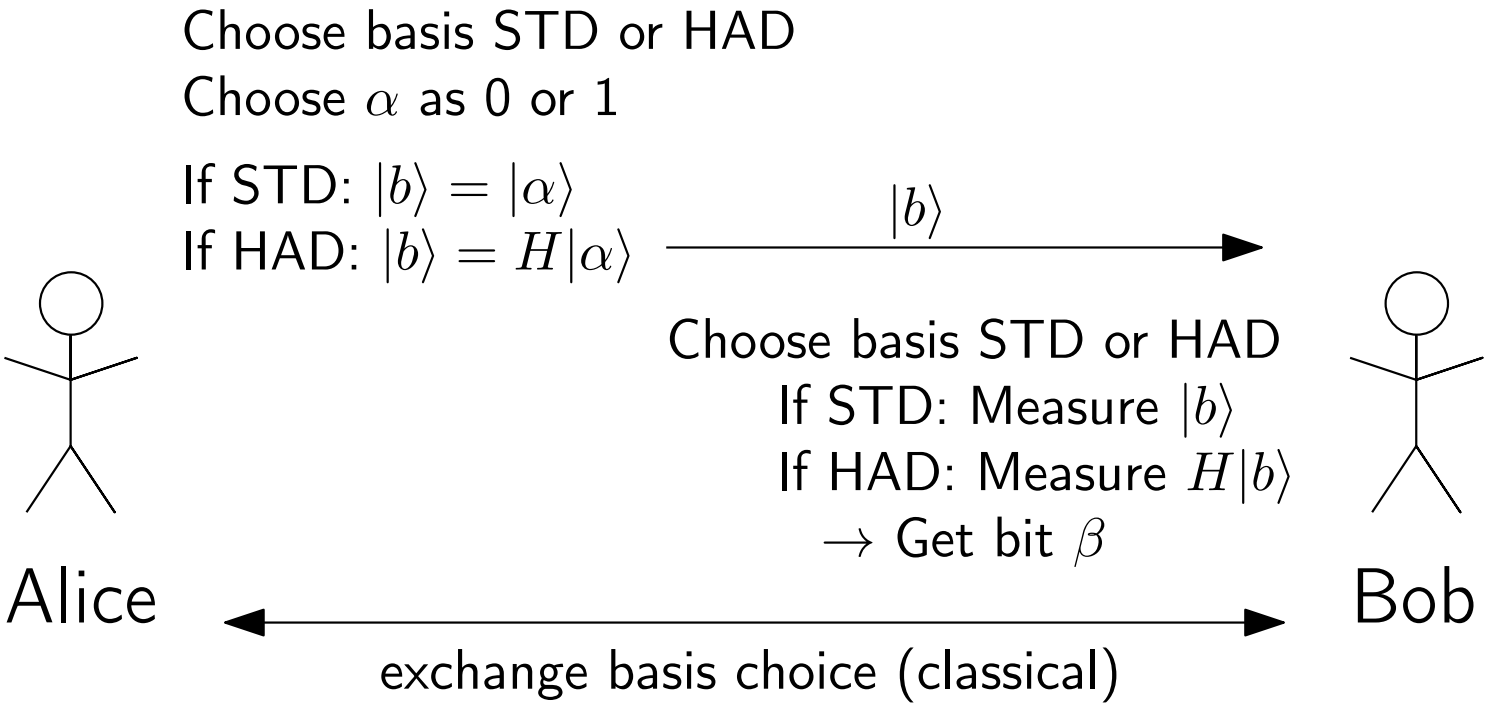
$$\xrightarrow{|b\rangle}$$

Choose basis STD or HAD
If STD: Measure $|b\rangle$
If HAD: Measure $H|b\rangle$
$\rightarrow$ Get bit β

Alice

Bob

Quantum Key Distribution

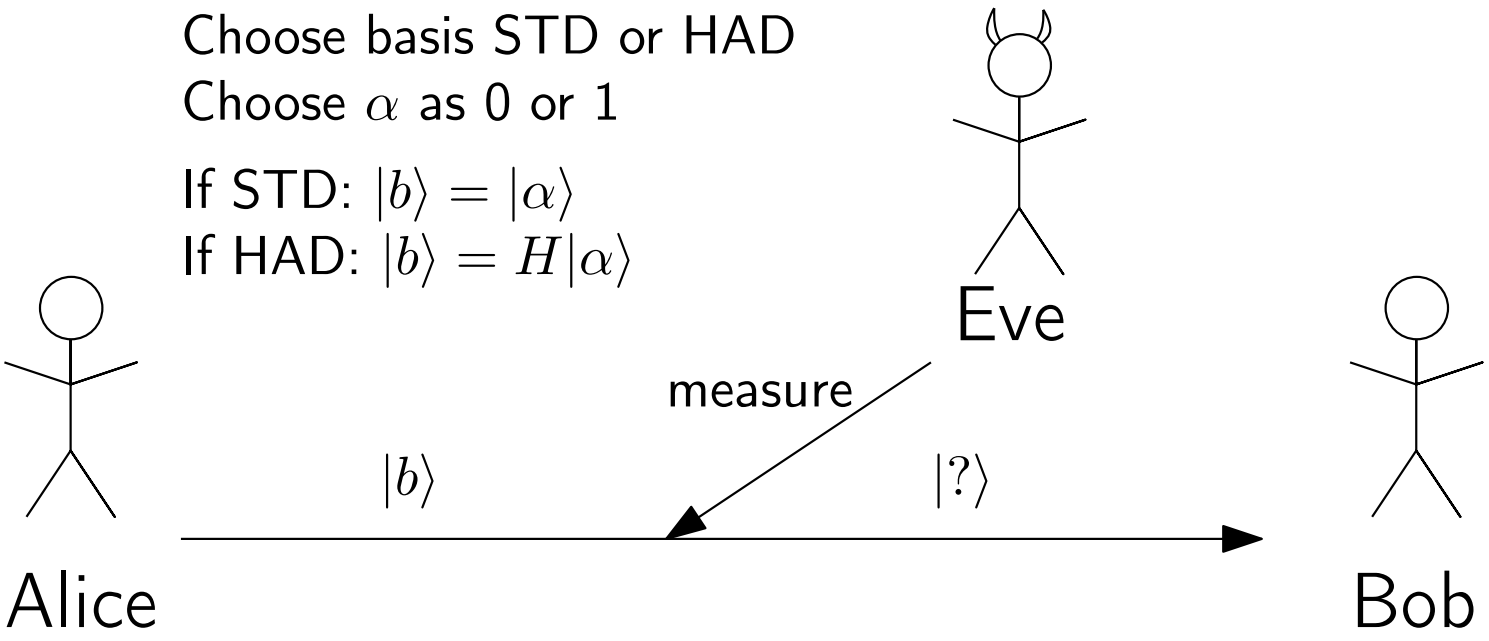


If they chose the same basis $\Rightarrow \alpha = \beta$

What can Eve do?

Choose basis STD or HAD
Choose α as 0 or 1

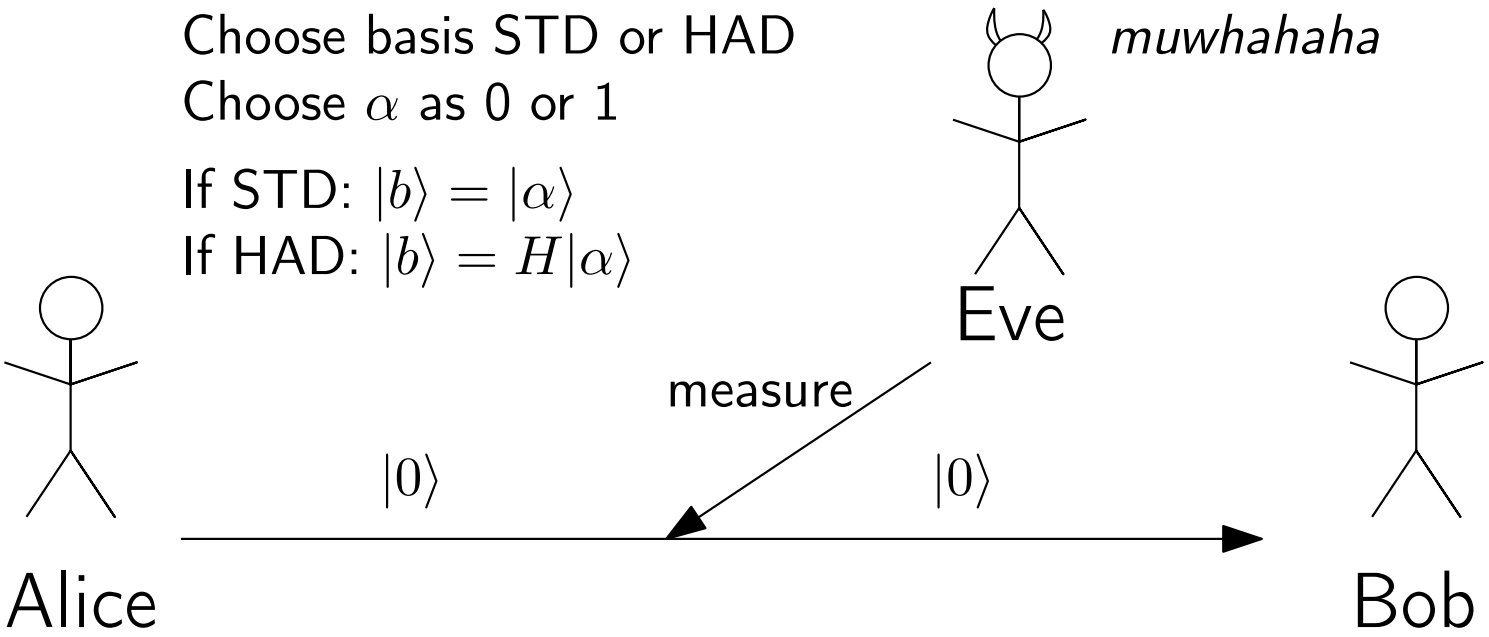
If STD: $|b\rangle = |\alpha\rangle$
If HAD: $|b\rangle = H|\alpha\rangle$



What can Eve do?

Choose basis STD or HAD
Choose α as 0 or 1

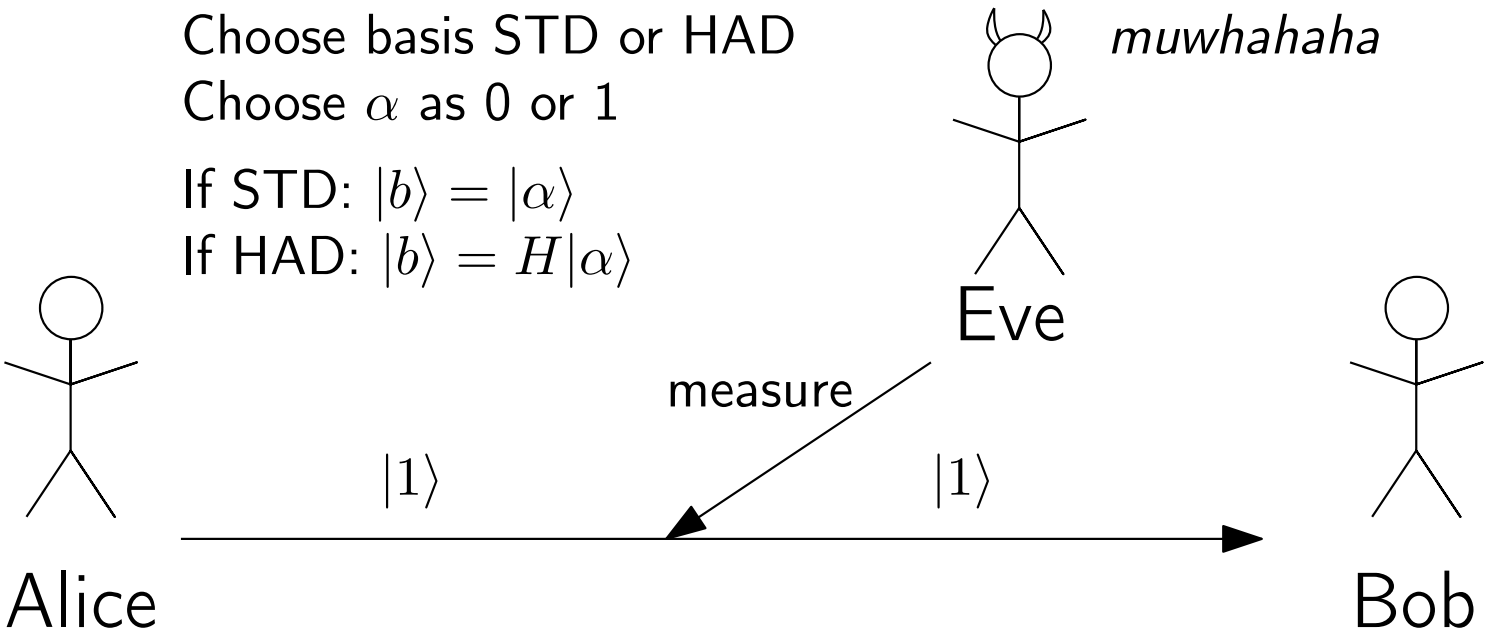
If STD: $|b\rangle = |\alpha\rangle$
If HAD: $|b\rangle = H|\alpha\rangle$



What can Eve do?

Choose basis STD or HAD
Choose α as 0 or 1

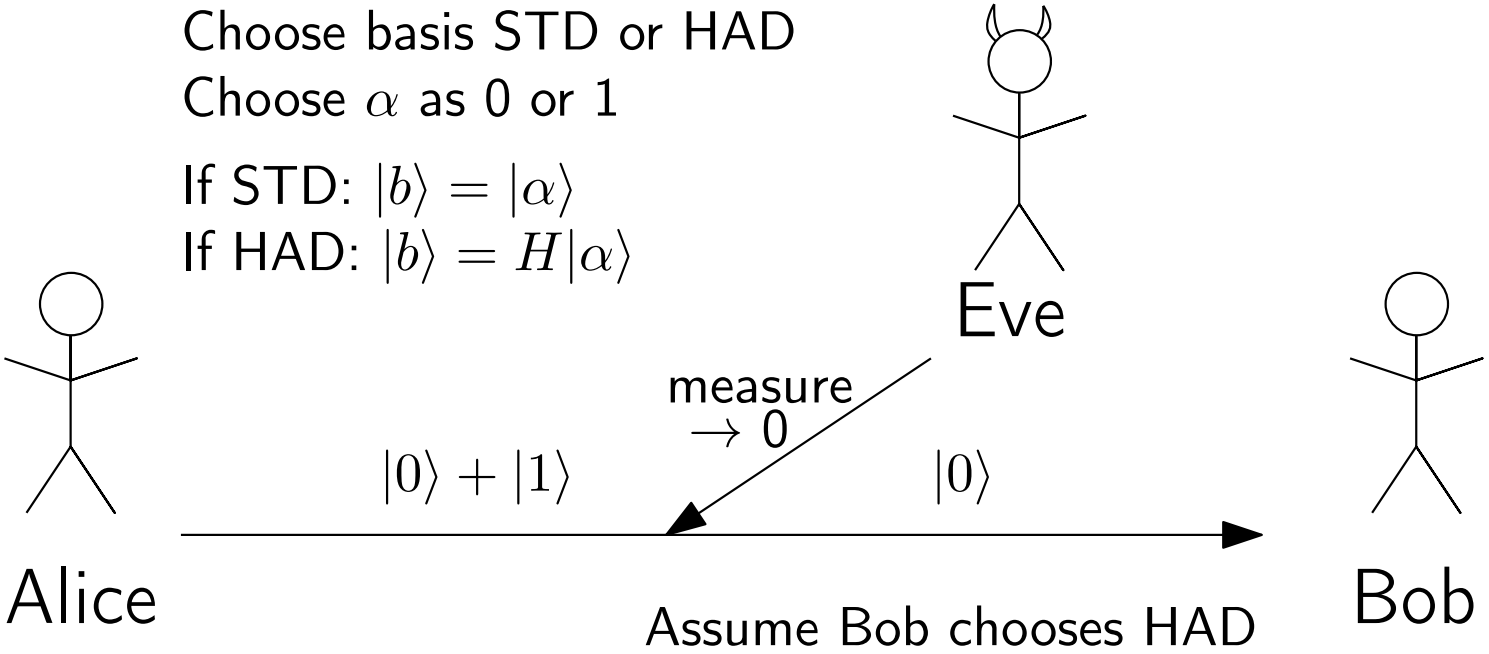
If STD: $|b\rangle = |\alpha\rangle$
If HAD: $|b\rangle = H|\alpha\rangle$



What can Eve do?

Choose basis STD or HAD
Choose α as 0 or 1

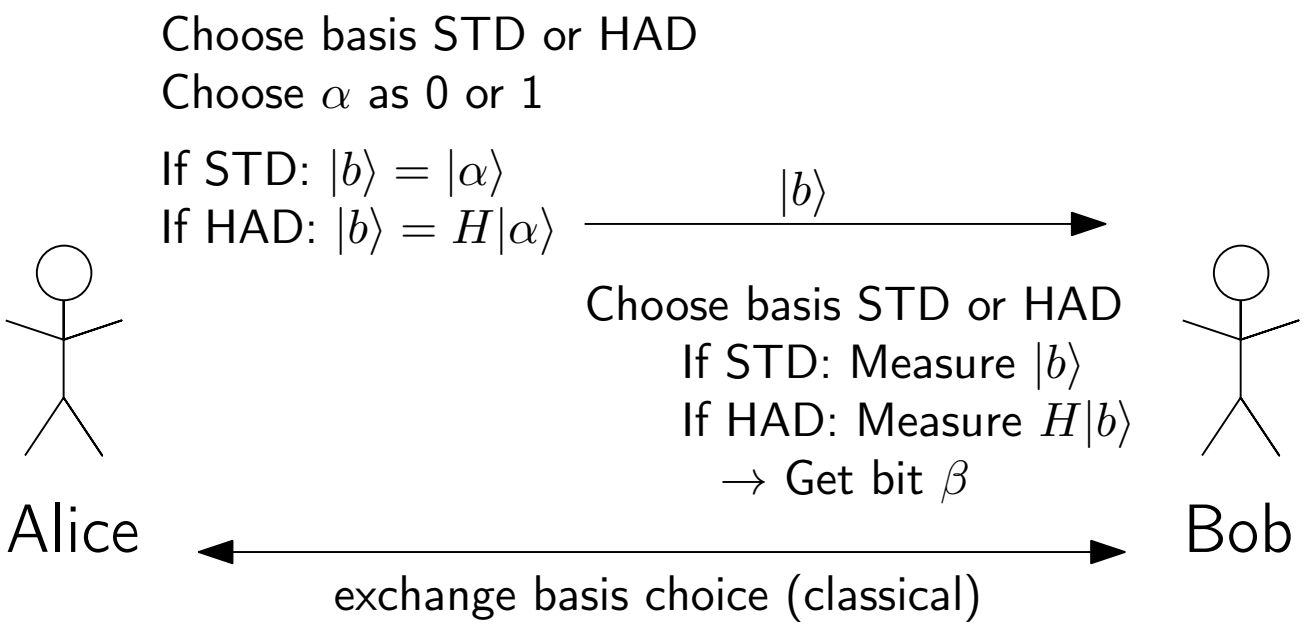
If STD: $|b\rangle = |\alpha\rangle$
If HAD: $|b\rangle = H|\alpha\rangle$



Assume Bob chooses HAD

measure $H|0\rangle = |0\rangle + |1\rangle$
$\rightarrow$ probability $\frac{1}{2}$: $\beta \neq \alpha$

How to detect Eve?



- If chosen different basis: Ignore this round
- Otherwise, $\alpha = \beta$
  ... unless Eve interfered!

But how can we detect when $\alpha \neq \beta$ without publishing it?

Multiple Qubits

If we have multiple qubits, we "multiply" them (this is a tensor product)

Examples

$$|0\rangle|0\rangle$$
$$|0\rangle|1\rangle|0\rangle|0\rangle|1\rangle...$$
$$|0\rangle(|0\rangle + |1\rangle) = |0\rangle|0\rangle + |0\rangle|1\rangle$$

"Entangled states"

$$|0\rangle|0\rangle + |1\rangle|1\rangle$$

Measurement and entangled states

Consider the "Bell state"

$$|0\rangle|0\rangle + |1\rangle|1\rangle$$

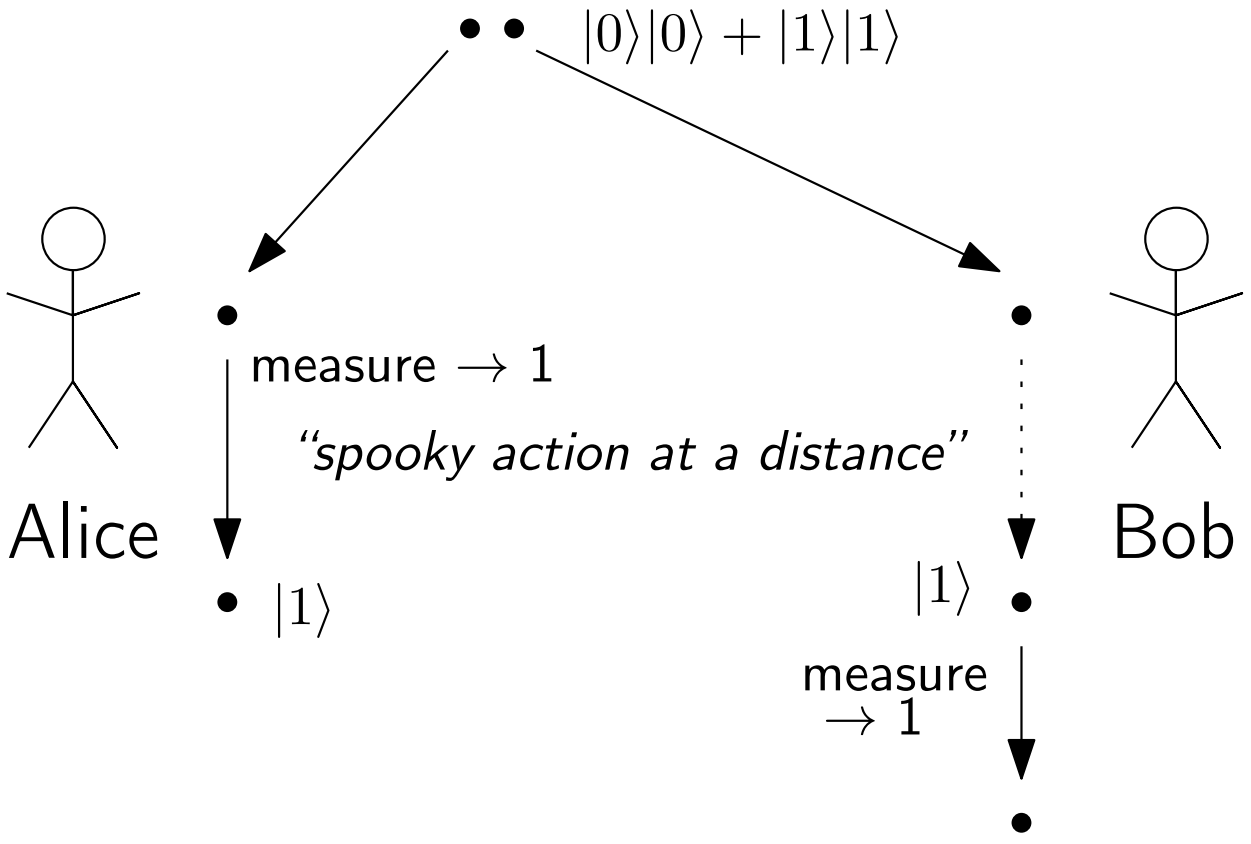
If we measure both qubits:

- Get $(0, 0)$ resp. $(1, 1)$ with equal probability
- New state is $|0\rangle|0\rangle$ resp. $|1\rangle|1\rangle$

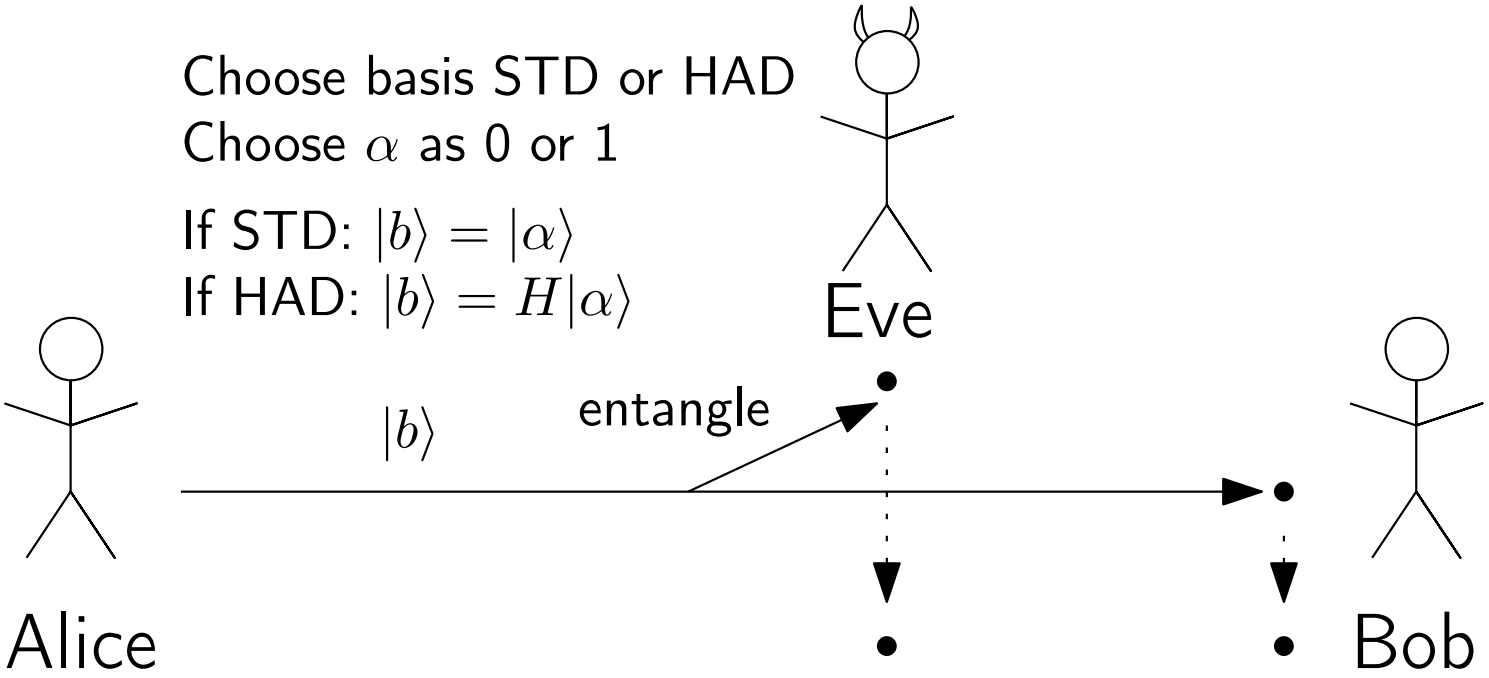
If we measure first qubit:

- Get 0 resp. 1 with equal probability
- New state is still $|0\rangle|0\rangle$ resp. $|1\rangle|1\rangle$!

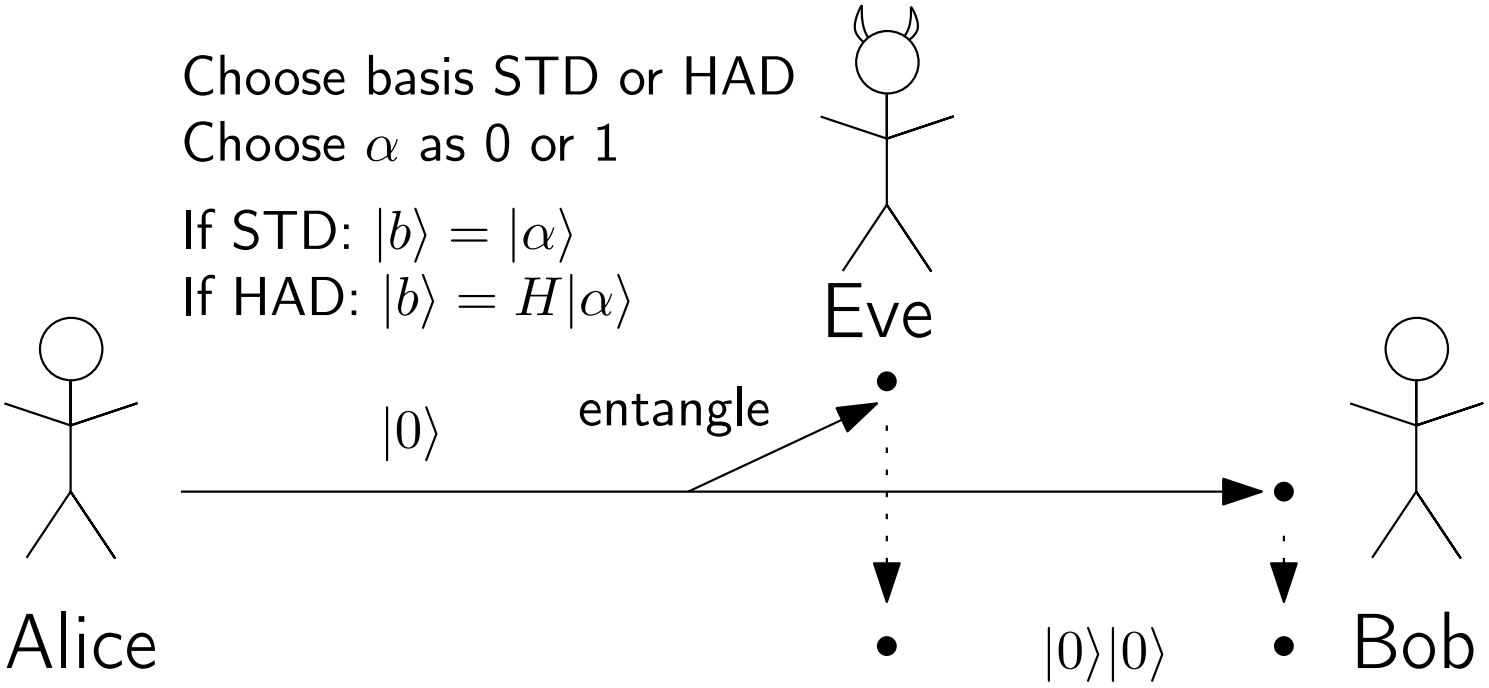
Measurement and entangled states



An Entanglement Attack

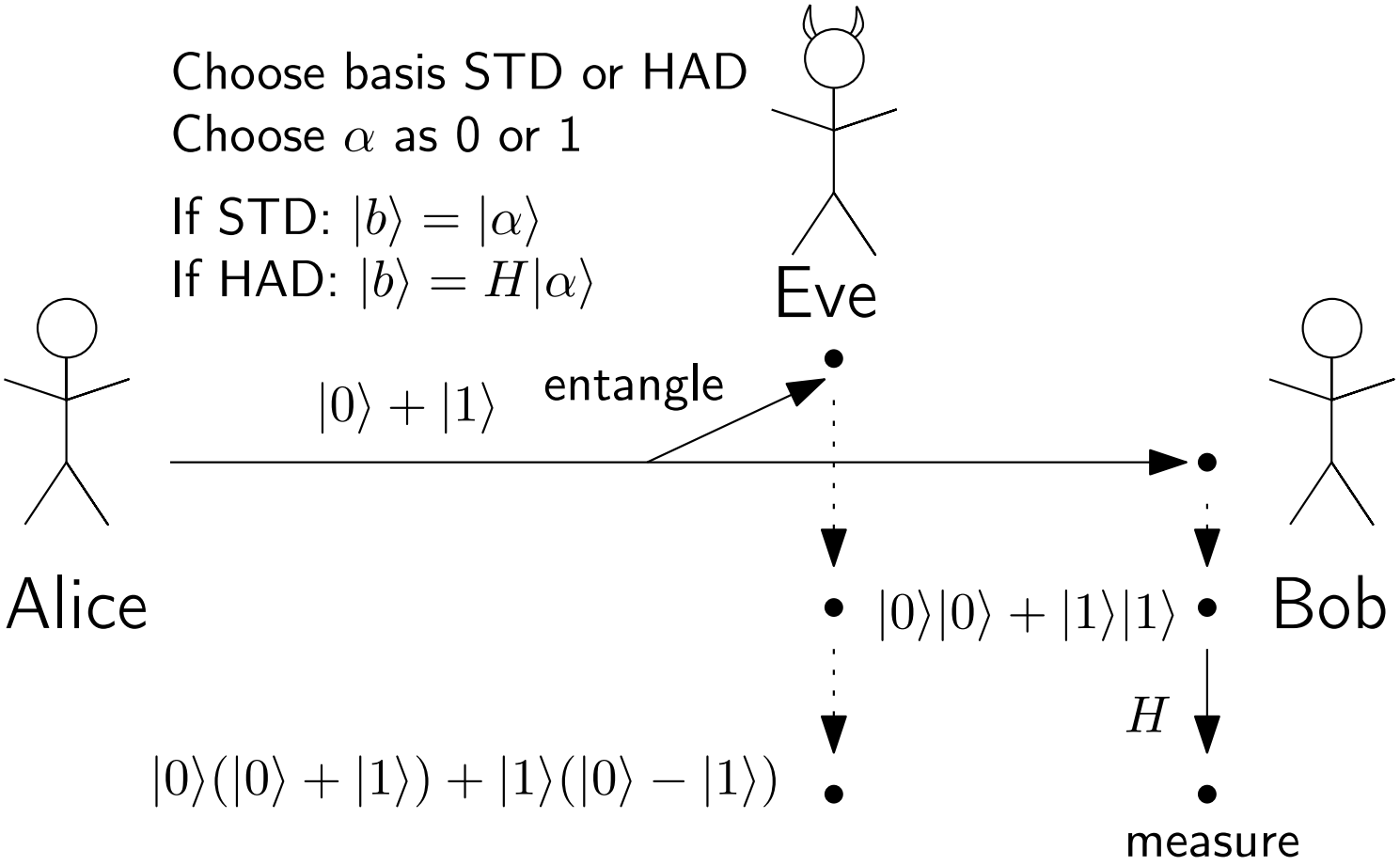


An Entanglement Attack

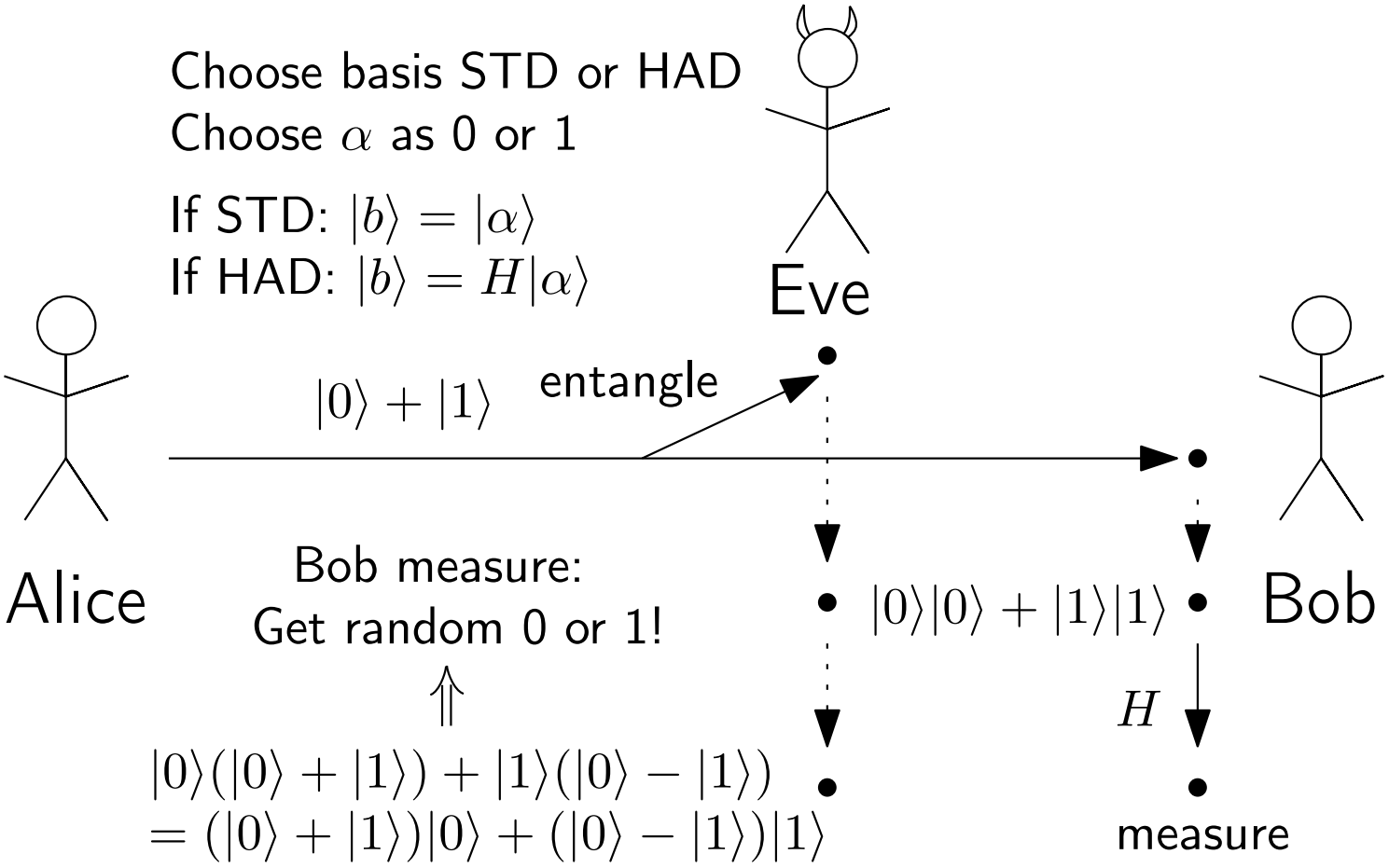


Bob measures in STD
Alice & Bob publish STD
$\Rightarrow$ Eve measures in STD!

An Entanglement Attack



An Entanglement Attack



What other attacks might be possible?

Choose basis STD or HAD
Choose α as 0 or 1

If STD: $|b\rangle = |\alpha\rangle$
If HAD: $|b\rangle = H|\alpha\rangle$

$$\xrightarrow{\quad |b\rangle \quad}$$

Choose basis STD or HAD
If STD: Measure $|b\rangle$
If HAD: Measure $H|b\rangle$
$\rightarrow$ Get bit β

Alice

Bob

$$\longleftrightarrow$$

exchange basis choice (classical)

Does this work in reality?

We do not need a quantum computer, there is only a single qubit!

Clavis XG QKD System

Quantum Key Distribution for production environments requiring high key transmission rate or extended range interconnection

- Long range (up to 150 km)
- Higher key rate on short distances: typical 14'000 AES-256 Keys per hour @ 24 dB
- Complex network topologies (ring, hub and spoke, meshed, star)
- Controlled and monitored centrally
- Interoperability with major Ethernet and OTN encryptors

However...

NCSC Position

Given the specialised hardware requirements of QKD over classical cryptographic key agreement mechanisms and the requirement for authentication in all use cases, the **NCSC does not endorse the use of QKD for any government or military applications**, and cautions against sole reliance on QKD for business-critical networks, especially in Critical National Infrastructure sectors.

www.ncsc.gov.uk/whitepaper/quantum-security-technologies; Accessed 06/11/2023

Discussion

- No hardness assumption
- It is cool

- Specialised hardware needed
- Hard to implement
- Classical authenticated channel

Other Quantum Cryptography

No-cloning theorem $\Rightarrow$ Unforgeability!

- Quantum money
- Copy-protected software

Cryptographic primitives under weaker assumptions

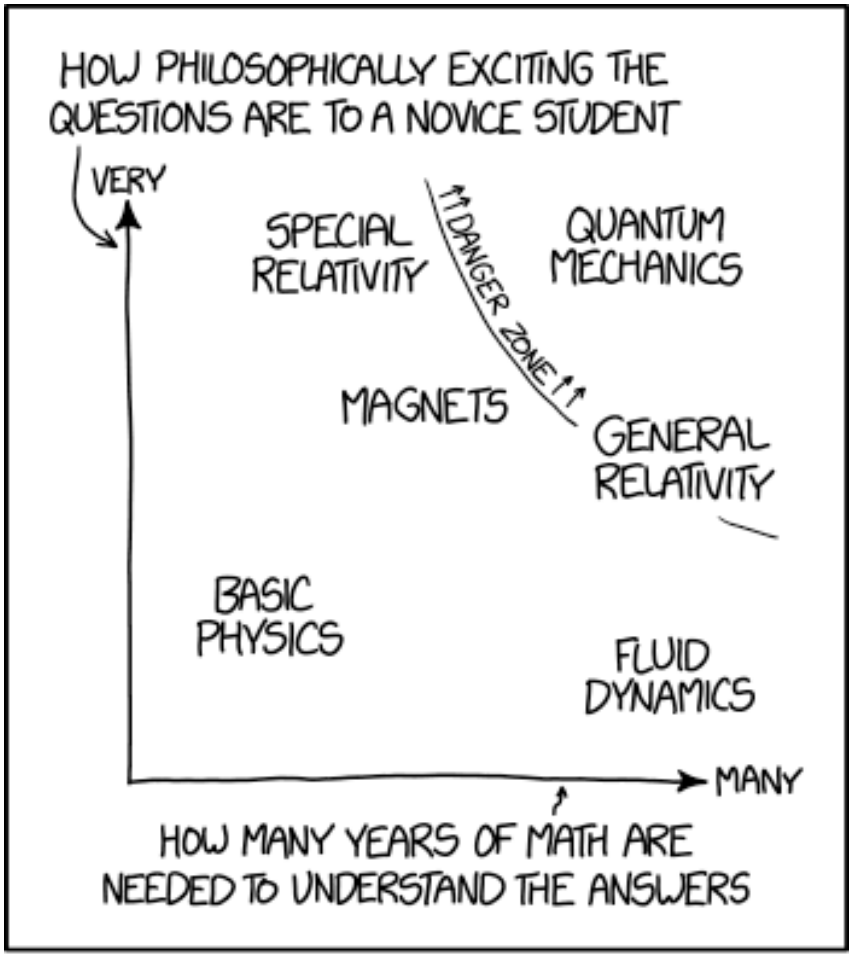
- Commitments
- Oblivious transfer and MPC
- Zero-knowledge proofs

Quantum varianss of cryptographic protocols

- Creating a secure *quantum* channel

Subject to ongoing research

Thank you for your attention!



WHY SO MANY PEOPLE HAVE WEIRD IDEAS ABOUT QUANTUM MECHANICS

www.xkcd.com/1861/